Dishita Chaturvedi

+91 7408470002 | dishitachaturvedi2005@gmail.com | Linkedin | Github

Summary

Driven AI/ML Engineering student eager to develop innovative solutions, with expertise in deep learning, predictive modeling, and tabular data classification. My experience includes impactful academic projects where I focused on ML model development and integration for areas like drug discovery, crop recommendation, and power consumption analysis. I'm skilled in Python, TensorFlow, PyTorch, C++, and have some familiarity with Java, complemented by strong problem-solving, research, collaboration, leadership, work assigning, and planning abilities. I'm actively looking for a dynamic developer role to apply my skills and create impactful AI applications.

Technical Skills

Programming Languages: Python, C++, SQL, Java

Machine Learning Learnings: Predictive Modeling, Statistical Analysis, Algorithms

Libraries & Tools: NumPy, Pandas, Scikit-learn, OpenCV, NLTK, Git, Docker

Projects

Power Consumption Analysis — May 2025 – July 2025
Machine Learning Project — *Python, Scikit-learn, Pandas, TensorFlow*

- Developed an end-to-end system for analyzing and optimizing household power consumption patterns
- Implemented Logistic Regression and Random Forest models for energy usage classification and prediction
- Achieved 86.2 percent accuracy in identifying peak consumption periods and inefficient appliances
- Integrated weather data and time-of-use pricing for personalized energy-saving recommendations
- Designed interactive dashboards for visualizing consumption trends and cost-saving opportunities

Crop Recommendation System — Feb 2025 – April 2025
Full-Stack ML Project — *Python, Flask, MongoDB, JavaScript*

- Developed an end-to-end system that recommends optimal crops based on geolocation and environmental factors
- Implemented machine learning models to analyze soil nutrients (N,P,K), rainfall, and climate data
- Built RESTful APIs with Flask for prediction service and Node.js for environmental data
- Designed interactive frontend with OpenWeather API integration for real-time location-based recommendations

Drug Discovery Model — Aug 2024 – Oct 2024
ML Research Project — *Python, TensorFlow, PyTorch, RDKit*

- Developed a predictive ML model to accelerate drug discovery by analyzing molecular bioactivity using Graph Neural Networks
- Implemented molecular graph representations (SMILES to graphs) to capture structural relationships for accurate target interaction predictions
- Achieved 87.9 percent accuracy in bioactivity classification, outperforming baseline models

Education

- **VIT Bhopal University, Bhopal** — Sep 2023 – Present
  Pursuing Bachelor of Technology in Computer Science (AI & ML Specialization)
  CGPA: 8.97 / 10
- **Delhi Public School, Varanasi** — 2018 – 2023
  12th Grade: 89% (2023)
  10th Grade: 94% (2021)

Certifications

- Microsoft Certified Azure Data Fundamentals
- Machine Learning in Partnership with Google Certificate